Isolated intracranial hypertension developing in SMA receiving nusinersen: A case report

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Case Report

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Abstract

In patients with spinal muscular atrophy (SMA) headache and vomiting after intrathecal administration of nusinersen is usually attributed to post-lumbar puncture syndrome. Additional adverse effects include idiopathic intracranial hypertension and hydrocephalus, with symptomatic intracranial hypertension also reported in adults. We describe a child with SMA type 2 who developed elevated lumbar puncture opening pressure (LOP) after the 12th injection of nusinersen, persisting at the 14th injection, without symptoms of increased LOP. Evaluations, including brain MRI, MRA, visual acuity, intraocular pressure, fundoscopy, revealed no abnormalities. This case suggests the possibility of isolated intracranial hypertension as a complication of repeated nusinersen therapy. Routine LOP monitoring is advised for long-term recipients to detect early complications.

Background

Spinal muscular atrophy (SMA) is an autosomal recessive neuromuscular disease caused by insufficient levels of survival motor neuron (SMN) protein[1, 2]. This deficiency leads to the loss of anterior horn motor neurons in the spinal cord, affecting the motor neurons within, resulting in progressive muscle atrophy and weakness[1, 3]. Nusinersen, the first antisense oligonucleotide used to treat SMA, is administered intrathecally and enhances the inclusion of exon 7 in SMN2, thereby elevating SMN protein levels[4, 5]. As nusinersen does not penetrate the blood-brain barrier, it requires intrathecal administration. The most prevalent side effects of nusinersen treatment, including headache and vomiting, are primarily associated with post-lumbar puncture syndrome[6].

Repeated nusinersen administration may increase intracranial pressure. Zhuang et al reported 5,324 nusinersen-treated cases, 196 adverse events (AEs) were documented, including 4 idiopathic intracranial hypertension cases with limited clinical details[7]. Rare AEs such as hydrocephalus, elevated lumbar puncture opening pressure (LOP), and papilledema have been observed in both pediatric and adult patients[8, 9]. A study of patients aged 0.7 to 20 years receiving intrathecal nusinersen found 35.3% had LOP > 28 cmH_2O (normal range: 10–25 cmH_2O), with a maximum of 50 cmH_2O. Only one patient reported transient headache. Two asymptomatic patients exhibited elevated LOP prior to treatment initiation. Data were limited to the first 10 injections, and intracranial pressure monitoring was inconsistent[10].

We report a case of a pediatric patient with SMA who developed isolated intracranial hypertension after 11 doses of nusinersen treatment, yet exhibited no clinical manifestations of elevated intracranial pressure.

Case presentation

The 10-year-old male first showed symptoms of delayed motor development at an age of 6 months. He was able to sit without support but soon lost the ability. He was never able to stand or walk unassisted.

At the age of 11 months the diagnosis of 5q-associated SMA was genetically confirmed. The patient has a homozygous deletion of exons 7 and 8 of the SMN1 gene and 3 copies of the SMN2 gene. Following alternating inpatient and home-based physical rehabilitation, the patient initially exhibited halted motor function regression with gradual improvement, achieving assisted standing as the highest motor milestone. However, due to the absence of disease-modifying therapies and inconsistent rehabilitation adherence, progressive motor decline emerged with age, predominantly affecting lower extremities.

During the initial evaluation at Xiangya Hospital in January 2022, the patient demonstrated independent sitting capacity, preserved writing ability, laborious rolling maneuvers, electric wheelchair mobility via joystick control, and loss of assisted standing/ambulation, alongside documented scoliosis and bilateral knee/ankle contractures. From January 2022 to May 2024, the patient received 11 intrathecal nusinersen doses without sedation. The first 7 doses yielded mild upper limb/trunk strength enhancement, improved handwriting fluency, independent rolling capacity, and score increases of 9 points on both the Hammersmith Functional Motor Scale Expanded and Revised Upper Limb Module. Subsequent doses 8–11 coincided with a 5 kg weight gain exacerbating rolling difficulty and recurrent pneumonia episodes necessitating rehabilitation discontinuation, culminating in gradual functional deterioration. The LOP was measured at 100 mmH_2O during the initial treatment. Throughout the first 11 treatment sessions, LOP fluctuated between 100–150 mmH_2O.

At the 12th dose, LOP was measured at 255 mmH_2O (normal range 40 ~ 180 mmH_2O), despite the absence of characteristic symptoms such as headache, visual disturbances, or vomiting, and normal cerebrospinal fluid parameters including cell count, protein, and glucose levels, the guardians declined further diagnostic investigations to elucidate the etiology of high LOP. At the 13th dose, LOP escalated to 290 mmH_2O, prompting mannitol administration despite persistent guardian refusal of diagnostic investigations. During the hospitalization for the 14th intrathecal nusinersen administration, a series of mandated neuro-ophthalmological evaluations encompassing brain MRI, MRA, fundoscopic examination, intraocular pressure measurement, and visual acuity testing were conducted, all demonstrating normal results. However, the LOP remained persistently elevated at 250 mmH_2O. Cerebrospinal fluid analysis (5 mL collected) demonstrated acellular fluid (0 cells/L) with biochemical parameters within normal ranges: glucose 3.69 mmol/L, lactate dehydrogenase 14.0 U/L, and protein 0.21 g/L. Given the patient's persistent intracranial hypertension lasting 8 months without any clinical symptoms, close observation with regular LOP monitoring was recommended, and pharmacological intervention was withheld at this stage.

Discussion

Our patient exhibited persistently elevated LOP without typical symptoms, meeting diagnostic criteria for isolated intracranial hypertension. Notably, this observation of asymptomatic LOP elevation in pediatric SMA patient undergoing nusinersen therapy may indicate a potential treatment-related phenomenon warranting further mechanistic investigation.

In the cohort reported by Becker et al., 12 patients exhibited elevated LOP levels; however, these data do not sufficiently establish a causal link to nusinersen[10]. Two patients had pre-existing LOP elevation prior to nusinersen administration. Three patients had incomplete LOP documentation, limited to a single elevated measurement. Other patients demonstrated fluctuating LOP levels, with transient isolated elevations during serial monitoring or rapid normalization after elevation[10]. Repeated lumbar punctures rarely lead to increased intracranial pressure, and no secondary complications such as infection, hemorrhage, or sinus venous thrombosis were observed in this patient. The cohort study revealed an association between LOP and sedation levels[10]. But our patient did not receive sedatives throughout the course of treatment. Compared to adults, the inherent biological variability in children and the compensatory capacity of the ventricular system to accommodate small volumetric changes may predispose to elevated LOP. However, we routinely withdraw 5 mL of CSF prior to nusinersen administration to mitigate cerebrospinal fluid volume fluctuations. Notably, the patient exhibited elevated LOP preceding three consecutive nusinersen injections, with measurements taken at four-month intervals.

Patients with SMA exhibit a significantly higher incidence of hydrocephalus than controls. Several case studies have documented hydrocephalus in SMA patients, primarily among those with severe phenotypes (type 0 and type 1)[11–13]. Surveillance bias may contribute to the observed increased risk, as SMA patients often undergo routine MRI scans that detect asymptomatic cases. This detection bias appears particularly relevant for adult patients, who frequently present with asymptomatic hydrocephalus[14]. Isolated cases of hydrocephalus during nusinersen treatment have been reported, though researchers have primarily attributed these events to the natural progression of SMA disease itself rather than drug-related adverse effects[8, 15]. In our patient, despite markedly elevated LOP, no clinical or imaging evidence of hydrocephalus was observed. This patient showed no clinical signs of intracranial hypertension, and ongoing monitoring remains essential for early detection of potential complications. Should symptoms like blurred vision, vomiting, or headache emerge, or if lumbar opening pressure measurements remain elevated without improvement, risdiplam administration could serve as a viable therapeutic option.

Conclusion

The etiology of this patient's isolated intracranial hypertension remains unclear, though the clinical timeline implicates elevated intracranial pressure as a possible complication of nusinersen therapy. Measuring lumbar opening pressure before every intrathecal injection remains essential, including during long-term nusinersen treatment. Additionally, post-procedural headaches, whether persistent or intermittent, should not be automatically attributed to post-lumbar puncture syndrome without thorough clinical evaluation.

Abbreviations

SMA
Spinal muscular atrophy
AEs
Adverse events
LOP
Lumbar puncture opening pressure
MRI
Magnetic Resonance Imaging
MRA
Magnetic Resonance angiography

Declarations

Data availability

All data are included in this study.

Authors' contributions

ZC and DH wrote the manuscript. XY collected clinical date. ZC, DH and PJ read and approved the final manuscript.

Ethical Approval

This study received expedited ethical approval from the Ethics committee of Xiangya Hospital, Central South University.

Consent to Participate

Written informed consent was obtained from both the patient and legal guardian, authorizing the use of clinical data for research purposes without influencing treatment decisions.

Consent for Publication

All authors approved the manuscript for publication. The patient’s father provided written consent for publication and submitted supplementary materials.

Competing interests

All authors have no conflict of interest to report.

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Clinical trial number

Not applicable.

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